HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Supported daily trading activities by providing data analytics and generating reports for a diverse portfolio of generation assets, including wind, solar, and battery storage.
- Developed and maintained analytical models for power price forecasting and market dynamics, documenting processes to ensure business continuity and best practice.
- Assisted in impact assessments and preparations for upcoming market changes, such as EirGrid's Future Power Markets programme, informing strategic trading decisions.
- Built and cultivated productive working relationships with external industry groups, including EirGrid and SEMO, facilitating information exchange on market dynamics.
- Collaborated with Commercial and Asset Management teams to evaluate and onboard new projects and operational assets, gaining cross-functional business experience.
- Communicated developments in the wholesale energy market to commercial and financial teams, delivering insights to inform strategic decision-making and maximise returns.
- Containerised market analysis tools using **Docker** and deployed them to an **AWS EC2** environment, ensuring reliable data provision for trading operations.

SKILLS

Analytics & Modelling – Power BI, Tableau, R, Python (pandas, NumPy, scikit-learn), SQL, DAX, Power Query, forecasting models, statistical analysis, optimisation

Energy Markets – Irish electricity market, renewable energy sector, wholesale electricity prices, market outcomes, I-SEM dynamics, Day-Ahead Market, Intraday Market, Balancing Market, commodity trading

Programming & Scripting – Python, Java, SQL, Spring Boot, REST APIs

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Business Tools – Microsoft Office Suite, Excel, Jira, Confluence

PROJECTS

Energy Market Forecasting Model

- Developed a Python-based forecasting model to predict wholesale electricity prices, utilising historical market data and identifying key influencing factors.
- Engineered data pipelines using **pandas** and **SQL** to extract, clean, and transform large datasets from multiple sources for model input.
- Visualised forecast results and model performance metrics using **Power BI**, presenting insights to stakeholders for strategic trading adjustments.

Renewable Asset Performance Analytics Dashboard

- Built an interactive dashboard in **Power BI** to monitor the operational performance of wind, solar, and battery assets, integrating data from various systems.
- Designed **DAX measures** and data models to calculate key performance indicators (KPIs) such as availability, energy output, and capacity factor, supporting commercial evaluation.
- Applied **Python** scripting for automated data ingestion and pre-processing, ensuring timely and accurate reporting for the trading team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)